

SECSR2043 OPERATING SYSTEMS
[20 Marks]

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 Section : 02

Marks

Instruction: Please answer all of the following questions. Whenever the 🖐 symbol appears, please raise your hand to call your instructor, he/she will verify your results by putting his / her initial next to the symbol.

1. Type the following commands using a text editor and save it as a *yourname.sh* (Example: *ahmad.sh*).

```
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt >
orange.txt
cd drinks; cp ../cars/*. *; cp ../fruits/*. *;
cp ../*.jar .
```

- a) Execute the script and draw a tree structure that contains created directories and files. The parent node of the directory begin with **\$HOME** directory.

[4 marks]



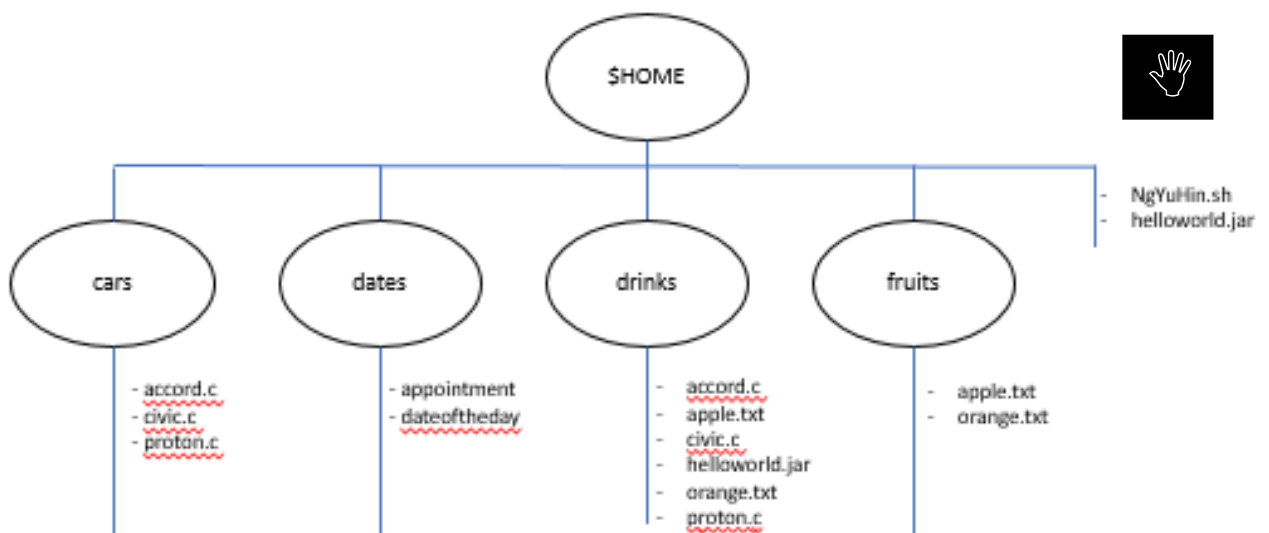
Print screen the script that you type;

Then draw the tree

```
rick@secr2043:~$ nano NgYuHin.sh
rick@secr2043:~$ chmod u+x NgYuHin.sh
rick@secr2043:~$ ./NgYuHin.sh
```

```
-- NgYuHin.sh
-- cars
  |-- accord.c
  |-- civic.c
  `-- proton.c
-- dates
  |-- appointment
  `-- dateoftheday
-- drinks
  |-- accord.c
  |-- apple.txt
  |-- civic.c
  |-- helloworld.jar
  |-- orange.txt
  `-- proton.c
-- fruits
  |-- apple.txt
  `-- orange.txt
```

```
-- helloworld.jar
```



- b) Write an interactive bash script that will read a type of file extension, display all those files, and count the number of files. To validate your script, display c program files, and enter “c” as the input to the bash script. [4 marks]

Print screen the bash script you type and run

```
rick@secr2043: ~  
GNU nano 7.2 interactivefiles.sh  
#!/bin/bash  
  
# Prompt the user to enter a file extension  
read -p "Enter the file extension (without the dot): " ext  
  
# Find and display all those files  
echo "Files with ${ext} extension:"  
find . -type f -name "*.${ext}"  
  
# Count the number of files  
count=$(find . -type f -name "*.${ext}" | wc -l)  
echo "Number of files with ${ext} extension: $count"  
rick@secr2043:~$ nano interactivefiles.sh  
rick@secr2043:~$ chmod u+x interactivefiles.sh  
rick@secr2043:~$ ./interactivefiles.sh  
Enter the file extension (without the dot): c  
Files with .c extension:  
./os_lab/echo.c  
./os_lab/sum.c  
./os_lab/hello.c  
./workspace/multiply.c  
./workspace/hello_world.c  
./workspace/kernel_info.c  
./workspace/argument.c  
./workspace/keyboard.c  
./workspace/input_keyboard.c  
./drinks/civic.c  
./drinks/accord.c  
./drinks/proton.c  
./subprocess1.c  
./subprocess2.c  
./cars/civic.c  
./cars/accord.c  
./cars/proton.c  
./example.c  
./mainprocess.c  
Number of files with .c extension: 19
```



2. The following Figure 1 illustrates a tree structure of some directories and files.

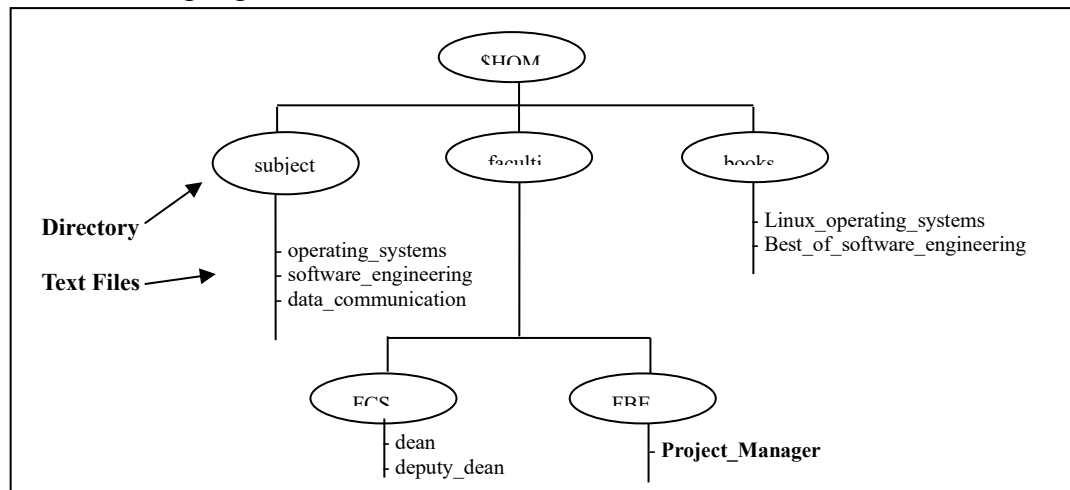


Figure 1

- a) Write a bash script (called `myname2a.sh`) that will produce directories and files as in Figure 1. Each text files contain its filename without the underscore character. For example: text file `Project_Manager` contains `Project Manager`). [4 marks]

Print screen the bash script you type and run

```

rick@secre2043: ~
GNU nano 7.2 NgYuHin2a.sh
#!/bin/bash

# Create directories
mkdir -p $HOME/subjects $HOME/faculties/FCS $HOME/faculties/FBE $HOME/books

# Create and write text files in subjects directory
echo "operating systems" > $HOME/subjects/operating_systems
echo "software engineering" > $HOME/subjects/software_engineering
echo "data communication" > $HOME/subjects/data_communication

# Create and write text files in faculties/FCS directory
echo "dean" > $HOME/faculties/FCS/dean
echo "deputy dean" > $HOME/faculties/FCS/deputy_dean

# Create and write text files in faculties/FBE directory
echo "Project Manager" > $HOME/faculties/FBE/Project_Manager

# Create and write text files in book directory
echo "Linux operating systems" > $HOME/books/Linux_operating_systems
echo "Best of software engineering" > $HOME/books/Best_of_software_engineering

rick@secre2043:~$ nano NgYuHin2a.sh
rick@secre2043:~$ chmod u+x NgYuHin2a.sh
rick@secre2043:~$ ./NgYuHin2a.sh
rick@secre2043:~$ tree $HOME
/home/rick
|-- NgYuHin.sh
|-- NgYuHin2a.sh
|-- books
|   |-- Best_of_software_engineering
|   |-- Linux_operating_systems
|-- cars
|   |-- accord.c
|   |-- civic.c
|   |-- proton.c
|-- dates
|   |-- appointment
|   |-- dateoftheday
|-- drinks
|   |-- accord.c
|   |-- apple.txt
|   |-- civic.c
|   |-- helloworld.jar
|   |-- orange.txt
|   |-- proton.c
|-- faculties
|   |-- FBE
|   |   |-- Project_Manager
|   |-- FCS
|   |   |-- dean
|   |   |-- deputy_dean
|-- fruits
|   |-- apple.txt
|   |-- orange.txt
|-- helloworld.jar
|-- interactivefiles.sh
|-- subjects
|   |-- data_communication
|   |-- operating_systems
|   |-- software_engineering

```



- [2 marks]

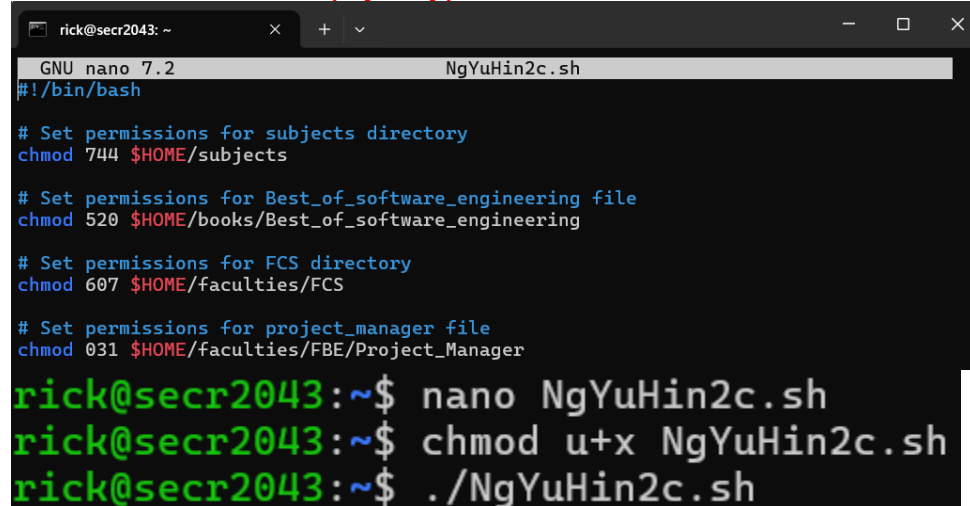


```
drwxr-xr-x 2 rick rick 4096 Jun 16 15:46 books
drwxr-xr-x 2 rick rick 4096 Jun 16 15:46 subjects
-rw-r--r-- 1 rick rick 29 Jun 16 15:46 Best_of_software_engineering
drwxr-xr-x 2 rick rick 4096 Jun 16 15:46 FCS
-rw-r--r-- 1 rick rick 16 Jun 16 15:46 Project_Manager
```

- [4 marks]

[illegible]

Print screen the bash script you type and run



```

rick@secr2043: ~
GNU nano 7.2 NgYuHin2c.sh
#!/bin/bash

# Set permissions for subjects directory
chmod 744 $HOME/subjects

# Set permissions for Best_of_software_engineering file
chmod 520 $HOME/books/Best_of_software_engineering

# Set permissions for FCS directory
chmod 607 $HOME/faculties/FCS

# Set permissions for project_manager file
chmod 031 $HOME/faculties/FBE/Project_Manager

rick@secr2043:~$ nano NgYuHin2c.sh
rick@secr2043:~$ chmod u+x NgYuHin2c.sh
rick@secr2043:~$ ./NgYuHin2c.sh

```

```

drwxr--r-- 2 rick rick 4096 Jun 16 15:46 subjects
-r-x-w---- 1 rick rick 29 Jun 16 15:46 Best_of_software_engineering
drw----rwx 2 rick rick 4096 Jun 16 15:46 FCS
-----wx--x 1 rick rick 16 Jun 16 15:46 Project_Manager

```

- d) Complete the following table by writing the access control for each directory or file after executing the bash script in question 2(c)). [2 marks]

Directory/File	Access Control
subjects	drwxr--r--
Best_of_software_engineering	-r-x-w----
FCS	drw----rwx
project_manager	-----wx--x

End of Lab 3

*** All the Best for Final Exam ***